

MindSpeak: A Real-Time BCI System for Silent Speech

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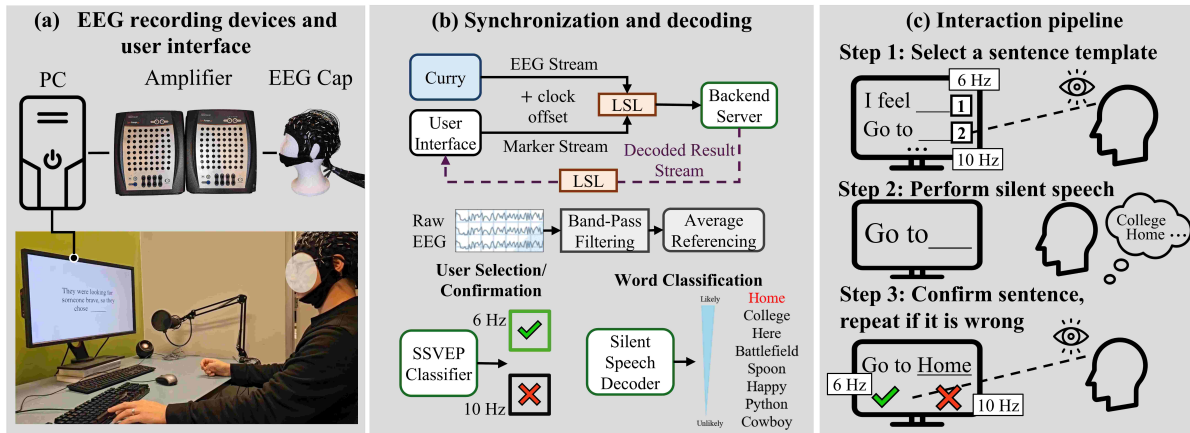


Figure 1: Overview of MindSpeak: a real-time BCI system for silent speech.

Abstract

This paper presents MindSpeak, a real-time brain-computer interface (BCI) system for recording, processing, and decoding silent speech to enable online multimodal communication between the human brain and a computer, involving both noninvasive multi-channel EEG signals and text output. To enable hand-free and brain-only control, our system incorporates steady-state visual evoked potential (SSVEP) for users to select incomplete sentences from a predefined pool and confirm the correctness of decoded words. An intuitive graphical interface is designed for natural communication. We evaluate the effectiveness of our real-time BCI system, which achieves 77.3% accuracy in decoding silent speech and 98.9% accuracy in SSVEP-based selection and confirmation of correct

sentences. Unlike existing BCI systems, the presented MindSpeak system significantly expands the application scope of existing BCI systems by enabling users to express complete thoughts through a fully BCI-controlled interactive interface. Our demonstration video is on: <https://youtu.be/B1wt1dmCCrg>.

CCS Concepts

• Theory of computation → Unsupervised learning and clustering; Social networks; • Human-centered computing → HCI design and evaluation methods; Natural language interfaces; Interaction techniques; • Computing methodologies → Natural language processing; Unsupervised learning.

Keywords

Real-time BCI system, Silent speech, EEG decoding

ACM Reference Format:

Jinzhao Zhou, Daniel Leong, Zehong Cao, Thomas Do, Sheng-Fu Liang, Tzyy-Ping Jung, and Chin-Teng Lin. 2025. MindSpeak: A Real-Time BCI System for Silent Speech. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25)*, October 27–31, 2025, Dublin, Ireland. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746027.3754486>

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MM '25, Dublin, Ireland

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ACM ISBN 979-8-4007-2035-2/2025/10
<https://doi.org/10.1145/3746027.3754486>

1 Introduction

Language is a natural and effective means of communication, but spoken speech isn't always possible, especially for individuals with impairments or in silent environments. To address this, we introduce MindSpeak, a real-time brain-computer interface (BCI) that enables silent speech communication using non-invasive EEG. Our system combines silent speech decoding with steady-state visually evoked potentials (SSVEP) to enable intuitive user interaction. Users select a sentence template via SSVEP, then silently articulate a target word. The EEG signals are decoded in real time to predict the word's meaning, which is inserted into the sentence. Users can then approve or reject the decoded word using SSVEP, ensuring accurate sentence construction.

2 System Framework

EEG recording devices: The devices used in the MindSpeak system is depicted in Figure 1-(a). Our MindSpeak system consist of a SynAmps RT amplifier, an 128-channel Neuroscan Quik-Cap, and a PC. The neuroscan amplifier sends electroencephalogram (EEG) data collected from the quik-cap at 1kHz sampling rate and is sent to the PC through the Curry 9 Neuroimaging software. The Curry software is set to push raw EEG signals to lsl stream while the backend server is implemented for all processing and event handling in the MindSpeak system.

Real-time Synchronization: Reliable Neuroscan-based BCI requires precise synchronization between GUI event markers and EEG streams from Curry. As shown in Figure 1-(b), EEG and markers are sent to the Lab Streaming Layer (LSL). To reduce misalignment that affects decoding, we estimate and correct clock offsets between streams. A multithreaded, non-blocking backend improves response time, achieving EEG-marker alignment within 25 ms. For real-time decoding, we accelerate the silent speech model using NVIDIA TensorRT [1].

Silent speech decoding: We built our silent speech decoder using a 12-word vocabulary from a prior EEG dataset [2], covering actions, emotions, locations, identity, and objects (e.g., Jumping, Happy, Home, Me, Spoon). This compact yet semantically diverse set supports effective communication in MindSpeak. We fine-tune a pretrained Large Brain Language Model (LBLM) [2] to classify these words using 1-second EEG segments, preprocessed with bandpass filtering and normalization. The model selects the word with the highest prediction score to complete sentence templates. To handle variability in user timing, we use time-shift augmentation during training.

SSVEP-based control: To enable full BCI control, we display SSVEP flickers on screen, each linked to a specific option. Users select by focusing on a flicker. The sentence selection interface (Figure 2-(a)) uses eight flickers (6, 6.5, 7, 7.5, 8, 8.5, 9, and 11 Hz), while the confirmation interface (Figure 2-(d)) uses two (6 Hz and 11 Hz) for accepting or rejecting outputs. SSVEP classification uses 2-second EEG from occipital/parietal channels (Pz, PO5, PO3, POz, PO4, PO6, O1, Oz, O2), processed with ms-CCA and ms-eTRCA [3, 4]. The classifier distinguishes 9 classes: 8 flicker frequencies and 1 no-action state.

3 Experiment results

To verify the effectiveness of the MindSpeak system, we conducted experiments on both silent speech decoding and SSVEP classification.

For silent speech decoding, we conducted cross-day evaluation by separating training, validation, and test sets by day. The fine-tuned LBLM was selected using validation performance and evaluated on a held-out test set. As shown in Table 2, our time-synchronization strategy notably improves accuracy, while time-shift augmentation reduces the generalization gap to just 0.5%. For SSVEP classification, our ms-CCA + ms-eTRCA method achieved up to 98.92% accuracy on the Tsinghua benchmark [5], outperforming popular alternatives and confirming its robustness for MindSpeak's selection and confirmation tasks.

Table 1: SSVEP Classification Performance

Method	Acc. (1s)	Acc. (2s)
ms-CCA + ms-eTRCA [3]	96.3± 1.6	98.9± 0.4
ms-CCA [4]	95.7± 1.7	98.7± 0.5
TDCA [6]	95.1± 2.5	98.9± 0.4

Table 2: Silent Speech Classification Performance

Sync.	T-Shift Aug.	Acc. Val ↑	Acc. Test ↑	Gen. Gap ↓
×	None	76.0	42.3	-33.7
✓	None	82.3	65.8	-16.5
✓	0.05	80.2	72.0	-8.2
✓	0.1	77.8	77.3	-0.5

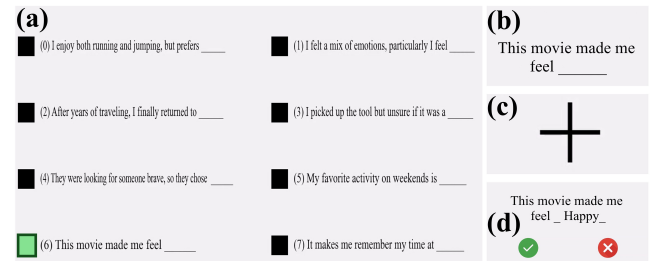


Figure 2: Details of User Interface.

4 User Experience

Figure 1-(c) and Figure 2 show the GUI pipeline. The user selects a sentence template via SSVEP flickers. Once selected, the sentence appears on screen, followed by a "+" cue prompting the user to silently articulate a target word. The decoded word is then inserted into the sentence.

Given that silent speech decoding is not always perfectly accurate, the interface includes a confirmation phase that lets users accept or reject the decoded sentence via green or red SSVEP flickers. This mirrors correction workflows in speech-to-text systems, supporting reliable and accessible communication, especially for users with speech impairments.

5 Acknowledgments

This work is partly supported by the Australian Research Council (ARC) under discovery grants DP210101093, DP220100803, DE220100265, DP250103612, and LP240200636. Research was also partially sponsored by the Australia Defence Innovation Hub under Contract No. P18-650825, Australian Cooperative Research Centres Projects (CRC-P) Round 11 CRCPXI000007, US Office of Naval Research Global under Cooperative Agreement Number ONRG - NICOP - N62909-19-1-2058, and AFOSR – DST Australian Autonomy Initiative agreement ID10134. We also thank the NSW Defence Innovation Network and the NSW State Government of Australia for financial support in part of this research through grant DINPP2019 S1-03/09 and PP21-22.03.02.

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